

Chapter 50 50.12a, 50.13, 50.17 eye structure, 50.24a, 50.26, 50.31 Adapted from Marieb, Elaine N; Hoehn, Katja, *Human Anatomy and Physiology*, 8th Ed., © 2010 Reprinted and electronically reproduced by Permission of Pearson Education, Inc., Upper Saddle River, New Jersey; **50.23** Data from K. L. Mueller et al., The receptors and coding logic for bitter taste, *Nature* 434:225–229 (2005); **50.35 grasshopper** Based on Hickman et al., *Integrated Principles of Zoology*, 9th ed., p. 518, Fig. 22.6, McGraw-Hill Higher Education, NY (1993); **Scientific Skills Exercise** Data from K. Schmidt-Nielsen, Locomotion: Energy Cost of Swimming, Flying, and Running, *Science* 177:222–228 (1972).

Chapter 51 51.4 Based on “*Drosophila*: Genetics Meets Behavior” by Marla B. Sokolowski, from *Nature Reviews: Genetics*, November 2001, Volume 2(11); **51.8** Data from *The Study of Instinct*, N. Tinbergen, Clarendon Press, Oxford (1951); **51.10** Adapted from “Prospective and Retrospective Learning in Honeybees” by Martin Giurfa and Julie Bernard, from *International Journal of Comparative Psychology*, 2006, Volume 19(3); **51.13** Adapted from Evolution of Foraging Behavior in *Drosophila* by Density Dependent Selection by Maria B. Sokolowski et al., from *Proceedings of the National Academy of Sciences USA*, July 8, 1997, Volume 94(14); **Scientific Skills Exercise** Data from Shell Dropping: Decision-Making and Optimal Foraging in Northwestern Crows by Reto Zach, from *Behaviour*, 1979, Volume 68(1–2); 51; **51.18** Reprinted by permission from Klaudia Witte; **51.24 Illustration** Adaptations of photograph by Jonathan Blair, Figure/PhotoID: 3.14, as appeared in *Animal Behavior: An Evolutionary Approach*, 8th Edition, Editor: John Alcock, p. 88. Reprinted by permission; **51.24 Map** data from “Rapid Microevolution of Migratory Behaviour in a Wild Bird Species” by P. Berthold et al., from *Nature*, December 1992, Volume 360(6405); **Art for Concept 51.2 Summary**: Data from *The Study of Instinct*, N. Tinbergen, Clarendon Press, Oxford (1951).

Chapter 52 52.19 Based on the data from *Ecology and Field Biology* by Robert L. Smith. Pearson Education, 1974; and *Sibley Guide to Birds* by David Allen Sibley. Random House, 2000; **52.20** Based on the data from W.J. Fletcher, Interactions among Subtidal Australian Sea Urchins, Gastropods and Algae: Effects of Experimental Removals, *Ecological Monographs* 57:89–109 (1987); **52.21** Based on S. D. Ling et al. Climate-Driven Range Extension of a Sea Urchin: Inferring Future Trends by Analysis of Recent Population Dynamics, *Global Change Biology* (2009) 15, 719–731, doi: 10.1111/j.1365-2486.2008.01734.x; **Scientific Skills Exercise** Based on the data from C. M. Crain et al., Physical and Biotic Drivers of Plant Distribution Across Estuarine Salinity Gradients, *Ecology* 85:2539–2549 (2004); **Graph for Test Your Understanding Question 11** Based on the data from J. Clausen et al., *Experimental Studies on the Nature of Species. III. Environmental Responses of Climatic Races of Achillea*, Carnegie Institution of Washington Publication No. 581 (1948).

Chapter 53 53.2 Data from A. M. Gormley et al., Capture-Recapture Estimates of Hector's Dolphin Abundance at Banks Peninsula, New Zealand, *Marine Mammal Science* 21:204–216 (2005); **Table 53.1** Data from P. W. Sherman and M. L. Morton, Demography of Belding's Ground Squirrel, *Ecology* 65:1617–1628 (1984); **53.4** Based on Demography of Belding's Ground Squirrels by Paul W. Sherman and Martin L. Morton, from *Ecology*, October 1984, Volume 65(5); **53.13** Data from Brood Size Manipulations in the Kestrel (*Falco tinnunculus*): Effects on Offspring and Parent Survival by C. Dijkstra et al., from *Journal of Animal Ecology*, 1990, Volume 59(1); **53.15** Based on Climate and Population Regulation: The Biogeographer's Dilemma by J. T. Enright, from *Oecologia*, 1976, Volume 24(4); **53.16** Based on the data from Predator Responses, Prey Refuges, and Density-Dependent Mortality of a Marine Fish by T.W. Anderson, *Ecology* 82(1):245–257 (2001); **53.18** Based on the data provided by Dr. Rolf O. Peterson; **53.21** Based on U.S. Census Bureau International Data Base; **53.22** Based on U.S. Census Bureau International Data Base; **53.23** Based on U.S. Census Bureau International Data Base; **53.24** Based on Ewing B., D. Moore, S. Goldfinger, A. Oursler, A. Reed, and M. Wackernagel. 2010. *The Ecological Footprint Atlas 2010*. Oakland: Global Footprint Network, p. 33 (www.footprintnetwork.org).

Chapter 54 54.1, 54.2 Based on A. Stanley Rand and Ernest E. Williams. The Anoles of La Palma: Aspects of Their Ecological Relationships, *Breviora*, Volume 327: 1–19. Museum of Comparative Zoology, Harvard University; **54.3** Data from J. H. Connell, The Influence of Interspecific Competition and Other Factors on the Distribution of the Barnacle *Chthamalus stellatus*, *Ecology* 42:710–723 (1961); **Scientific Skills Exercise** Based on the data from B. L. Phillips and R. Shine, An Invasive Species Induces Rapid Adaptive Change in a Native Predator: Cane Toads and Black Snakes in Australia, *Proceedings of the Royal Society B* 273:1545–1550 (2006); **54.11** Based on the data from Sally D. Hacker and Mark D. Bertness, Experimental Evidence for Factors Maintaining Plant Species Diversity in a New England Salt Marsh. *Ecology*, September 1999, Volume

80(6); **54.14 Graph** Data from N. Fierer and R. B. Jackson, The Diversity and Biogeography of Soil Bacterial Communities, *Proceedings of the National Academy of Sciences USA* 103:626–631 (2006); **54.17** Based on George A. Knox. Antarctic Marine Ecosystems, from *Antarctic Ecology*, Volume 1, edited by Martin W. Holdgate. Academic Press, 1970; **54.18** Adapted from Denise L. Breitburg et al., Varying Effects of Low Dissolved Oxygen on Trophic Interactions in an Estuarine Food Web. *Ecological Monographs*, November 1997, Volume 67(4). Used by permission of the Ecological Society of America; **54.19** Based on B. Jenkins et al., Productivity, Disturbance and Food Web Structure at a Local Spatial Scale in Experimental Container Habitats. *OIKOS*, November 1992, Volume 65(2); **54.20 Graph** Data from R. T. Paine, Food web complexity and species diversity, *American Naturalist* 100:65–75 (1966); **54.24** Based on the data from C.R. Townsend, M.R. Scarsbrook, and S. Doledet, The Intermediate Disturbance Hypothesis, Refugia, and Biodiversity in Streams, *Limnology and Oceanography* 42:938–949 (1997); **54.26** Based on Robert L. Crocker and Jack Major. Soil Development in Relation to Vegetation and Surface Age at Glacier Bay, Alaska. *Journal of Ecology*, July 1955, Volume 43(2); **54.27** Adapted from F. Stuart Chapin et al., Mechanisms of Primary Succession Following Deglaciation at Glacier Bay. *Ecological Monographs*, May 1994, Volume 64(2). Ecological Society of America; **54.29** Adapted from D. J. Currie. Energy and Large-Scale Patterns of Animal-and Plant-Species Richness. *American Naturalist*, January 1991, Volume 137(1): 27–49; **54.30** Adapted from Robert H. MacArthur and Edward O. Wilson, An Equilibrium Theory of Insular Zoogeography. *Evolution*, December 1963, Volume 17(4). Society for the Study of Evolution; **54.33** Based on Daniel S. Simberloff and Edward O. Wilson. 1969. Experimental Zoogeography of Islands: The Colonization of Empty Islands. *Ecology*, Vol. 50, No. 2 (Mar., 1969), pp. 278–296.

Chapter 55 55.4 Based on Figure 1.2 from Donald L. DeAngelis (1992), *Dynamics of Nutrient Cycling and Food Webs*. Taylor & Francis; **55.6** Data from J. H. Ryther and W. M. Dunstan, Nitrogen, Phosphorus, and Eutrophication in the Coastal Marine Environment, *Science* 171:1008–1013 (1971); **Table 55.1** Data from D. W. Menzel and J. H. Ryther, Nutrients Limiting the Production of Phytoplankton in the Sargasso Sea, with Special Reference to Iron, *Deep Sea Research* 7:276–281 (1961); **55.7** Based on the data from Fig. 4.1, p. 82, in R.H. Whittaker (1970), *Communities and Ecosystems*. Macmillan, New York; **55.8** Based on Fig. 3c and 3d from Temperate Forest Health in an Era of Emerging Megadisturbance, Constance I. Millar and Nathan L. Stephenson, *Science* 349, 823 (2015); doi: 10.1126/science.aaa9933; **Scientific Skills Exercise** Data from J. M. Teal, Energy Flow in the Salt Marsh Ecosystem of Georgia, *Ecology* 43:614–624 (1962); **55.12** Data from J. A. Trofymow and the CIDET Working Group, The Canadian Intersite Decomposition Experiment: Project and Site Establishment Report (Information Report BC-X-378), Natural Resources Canada, Canadian Forest Service, Pacific Forestry Centre (1998) and T. R. Moore et al., Litter decomposition rates in Canadian forests, *Global Change Biology* 5:75–82 (1999); **55.14** Adapted from Figure 7.4 from Robert E. Ricklefs (2001), *The Economy of Nature*, 5th edition. W.H. Freeman and Company; **55.18b** Based on the data from Wei-Min Wu et al. (2006), Pilot-Scale In Situ Bioremediation of Uranium in a Highly Contaminated Aquifer. 2. Reduction of U(VI) and Geochemical Control of U(VI) Bioavailability. *Environmental Science Technology* 40 (12):3986–3995 (5/13/06); **Art for Concept 55.1 Summary** Based on Figure 1.2 from Donald L. DeAngelis (1992). *Dynamics of Nutrient Cycling and Food Webs*. Taylor & Francis.

Chapter 56 56.10 Based on data from Gene Likens; **56.11** Krebs, Charles J., *Ecology: The Experimental Analysis of Distribution and Abundance*, 5th Ed., © 2001. Reprinted and electronically reproduced by permission of Pearson Education, Inc., Upper Saddle River, New Jersey; **56.13** Data from “Tracking the Long-Term Decline and Recovery of an Isolated Population” by R.L. Westemeier et al., *Science* Volume 282(5394):1695–1698 (11/27/98), AAAS; **56.19** Adapted from Norman Myers et al. (2000). Biodiversity Hotspots for Conservation Priorities, *Nature*, February 24, 2000, Volume 403(6772); **56.28** Based on CO₂ data from www.esrl.noaa.gov/gmd/ccgg/trends. Temperature data from www.giss.nasa.gov/gistemps/graphs/Fig. A.1rg.gif; **Scientific Skills Exercise** Based on data from National Oceanic & Atmospheric Administration, Earth System Research Laboratory, Global Monitoring Division; **56.32** Based on the data from “History of the Ozone Hole,” from NASA website, February 26, 2013; and “Antarctic Ozone,” from British Antarctic Society website, June 7, 2013; **56.35** Based on the data from Instituto Nacional de Estadística y Censos de Costa Rica and Centro Centroamericano de Población, Universidad de Costa Rica.

Appendix A Figure 5.11 Wallace/Sanders/Ferl, *Biology: The Science of Life*, 3rd Ed., © 1991. Reprinted and electronically reproduced by permission of Pearson Education, Inc., Upper Saddle River, New Jersey.